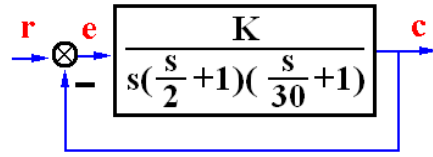


频率法串联超前校正

例 2 系统结构图如右图所示



指标要求：

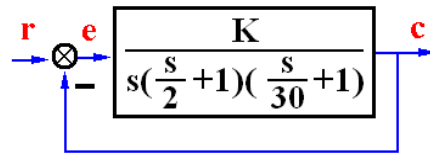
$$\begin{cases} r(t) = t, & e_{ss} \leq 0.1 \\ \sigma\% \leq 27.5\% \\ t_s \leq 1.7 \\ h^* \geq 10 \text{ dB} \end{cases} \xrightarrow[\text{5-56}]{\text{P173}} \begin{cases} \gamma^* \geq 50^\circ \\ t_s = \frac{8.5}{\omega_c^*} \leq 1.7 \end{cases} \Rightarrow \begin{cases} \omega_c^* \geq 8.5/1.7 = 5 \\ \gamma^* \geq 50^\circ \\ h^* \geq 10 \text{ dB} \end{cases}$$

试确定 $G_c(s) = ?$

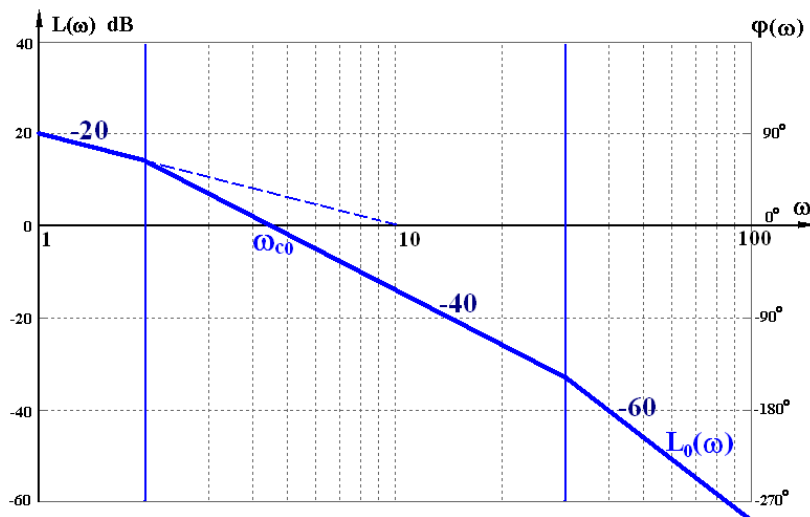
解： $G(s) = \frac{K}{s(\frac{s}{2} + 1)(\frac{s}{30} + 1)}$ $\begin{cases} \text{开环增益: } K \\ v = 1 \end{cases}$

① 由 e_{ss}^* 要求确定 K ：

$$e_{ss}^* = \frac{1}{K} \leq 0.1 \rightarrow K = 10$$



② 作 $L(\omega)$ 曲线如右



$$\omega_{c0} = \sqrt{2 \times 10} = 4.472 < 5$$

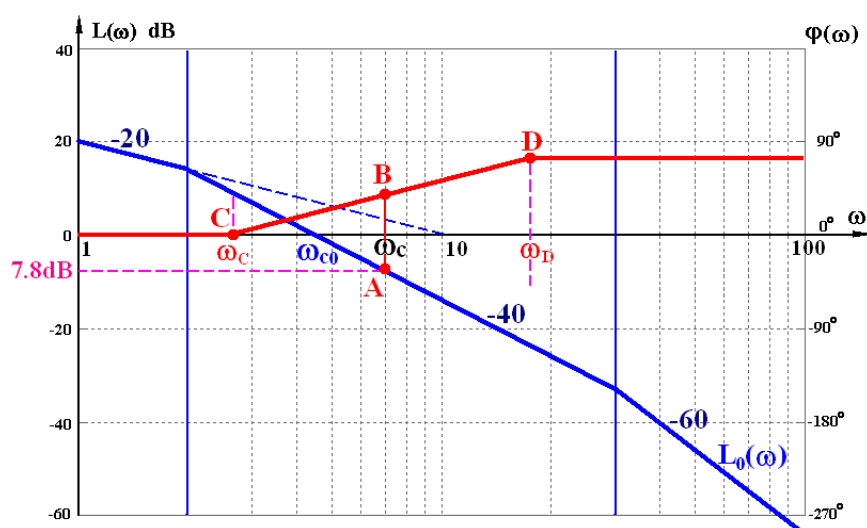
$$\begin{aligned} \gamma_0 &= 90^\circ - \tan^{-1} \frac{4.472}{2} - \tan^{-1} \frac{4.472}{30} \\ &= 90^\circ - 65.9^\circ - 8.5^\circ = 15.6^\circ < 50^\circ \end{aligned}$$

③ 采用超前校正：定超前网络参数：

$$\begin{aligned}\varphi_m &= \gamma^* - \gamma_0 + (5^\circ \sim 10^\circ) \\ &= 50^\circ - 15.6^\circ + 10^\circ = 44.4^\circ\end{aligned}$$

$$\begin{cases} a = \frac{1 + \sin \varphi_m}{1 - \sin \varphi_m} = 5.66 \approx 6 \\ 10 \lg a = 7.8 \text{ dB} \end{cases}$$

④ 确定 $G_c(s)$ $-7.8 \text{ dB} \rightarrow A, B, C, D$



$$7.8 \text{ dB} = 40 \cdot \lg \frac{\omega_c}{\omega_{c0}}$$

$$\omega_c = \omega_{c0} \cdot 10^{\frac{7.8}{40}} = 4.472 \times 1.567 = 7 > 5$$

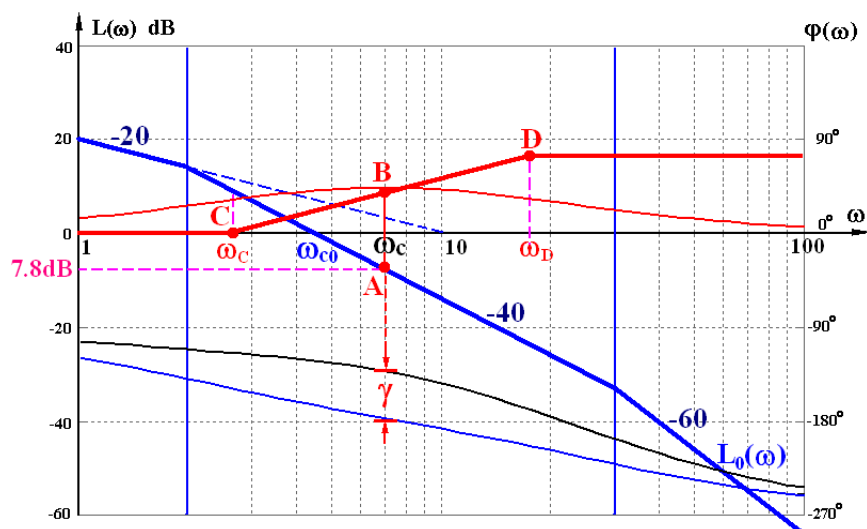
按 γ^* 设计

$$\omega_C = \frac{\omega_{c0}^2}{\omega_c} = \frac{4.47^2}{7} = 2.86$$

$$\omega_D = \frac{\omega_c^2}{\omega_C} = \frac{7^2}{2.86} = 17.13$$

$$G_{c1}(s) = \frac{\frac{s}{\omega_C} + 1}{\frac{s}{\omega_D} + 1} = \frac{\frac{s}{2.86} + 1}{\frac{s}{17.13} + 1}, \quad a = \frac{17.13}{2.86} = 6$$

$$G_1(s) = G_{c1}(s) \cdot G_0(s) = \frac{\frac{s}{2.86} + 1}{\frac{s}{17.13} + 1} \cdot \frac{10}{s(\frac{s}{2} + 1)(\frac{s}{30} + 1)}$$

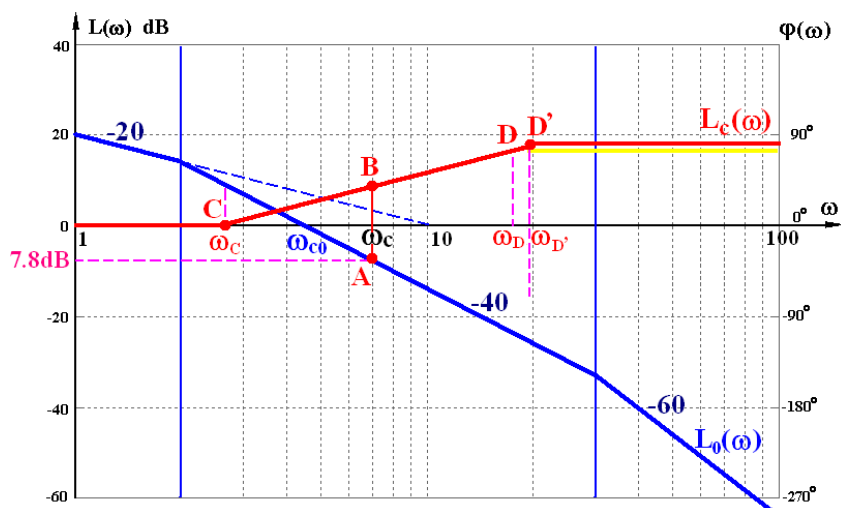


⑤ 验算: $K_v = K = 10$

$$\omega_c = 7 > 5$$

$$\begin{aligned} \gamma_1 &= 180^\circ + \arctan \frac{7}{2.86} - 90^\circ - \arctan \frac{7}{2} - \arctan \frac{7}{30} - \arctan \frac{7}{17.13} \\ &= 180^\circ + 67.78^\circ - 90^\circ - 74.05^\circ - 13.13^\circ - 22.22^\circ = 48.38^\circ < 50^\circ \end{aligned}$$

相角裕度差 1.62° , 适当延长 D 点, 调整 $\omega_{D'} = 19$ (其它参数不变)。

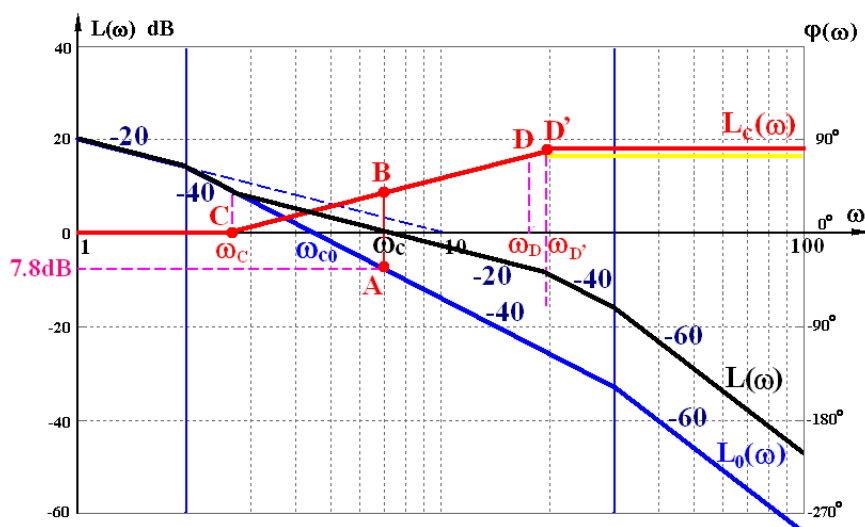


则
$$\begin{aligned} \gamma_2 &= \gamma_1 + \arctan \frac{7}{\omega_D} - \arctan \frac{7}{\omega_{D'}} \\ &= 48.38^\circ + 22.22^\circ - 20.22^\circ = 50.38^\circ > 50^\circ \end{aligned}$$

故选定

$$G_c(s) = \frac{\frac{s}{2.86} + 1}{\frac{s}{19} + 1}$$

$$G(s) = G_c(s) \cdot G(s) = \frac{\frac{s}{2.86} + 1}{\frac{s}{19} + 1} \cdot \frac{10}{s(\frac{s}{2} + 1)(\frac{s}{30} + 1)}$$



验算 h 指标：先找出 ω_g ，依图分析， $\omega_g \approx \sqrt{19 \times 30} = 23.875$

$$\angle G^*(\omega_g) = \arctan \frac{\omega_g}{2.86} - 90^\circ - \arctan \frac{\omega_g}{2} - \arctan \frac{\omega_g}{30} - \arctan \frac{\omega_g}{19}$$

$$\begin{aligned} \omega_g &= 23.875 \\ &= 83.169^\circ - 90^\circ - 85.212^\circ - 38.51^\circ - 51.49^\circ = -182.4^\circ \end{aligned}$$

$$\begin{aligned} \omega_g &= 23 \\ &= 82.91^\circ - 90^\circ - 85.05^\circ - 37.476^\circ - 50.44^\circ = -179.5^\circ \end{aligned}$$

$$\begin{aligned} \omega_g &= 23.1 \\ &= 82.94^\circ - 90^\circ - 85.05^\circ - 37.60^\circ - 50.56^\circ = -180.27^\circ \end{aligned}$$

$$\omega_g = 23.1$$

$$|G(\omega_g)| = \frac{1}{h} = \frac{\left[\left(\frac{23.1}{2.86} \right)^2 + 1 \right]^{\frac{1}{2}} \times 10}{23.1 \left[\left(\frac{23.1}{2} \right)^2 + 1 \right]^{\frac{1}{2}} \left[\left(\frac{23.1}{30} \right)^2 + 1 \right]^{\frac{1}{2}} \left[\left(\frac{23.1}{19} \right)^2 + 1 \right]^{\frac{1}{2}}} = \frac{1}{6.5}$$

$$h = 6.5$$

$$20 \lg h = 20 \lg 6.5 = 16.26 > 10 \text{ dB}$$

所以选 $G_c(s) = \frac{s}{\frac{s}{19} + 1} + 1$ ，可以满足要求。

